

Project Rubric – Section 1: The Finished Product60

Student Name: _____ Course/Section: _____ Date: _____

Category	<i>All must be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	Section	Score
Completeness of All Mandatory Sections					- Not all mandatory sections are complete. [STOP GRADING]	
Category	5	3	1	0	Section	Score
Functionality and assistance.	- Fully functional with successful sign-off after an initial review	- Failure of successful sign-off after an initial review.	- Failure of successful sign-off after a second review.	- Failure of successful sign-off after a third review.	Section A Mandatory	
	- Completed with no assistance	- First instance of assistance for this section.	- Second instance of assistance for this section.	- Third instance of assistance for this section.	Section B Optional	
	- Demonstrated a systematic approach to problem solving with a high degree of effectiveness and/or professionalism	- Demonstrated a systematic approach to problem solving with considerable effectiveness and/or professionalism	- Demonstrated a systematic approach to problem solving with some effectiveness and/or professionalism	- Demonstrated a systematic approach to problem solving with limited or no effectiveness and/or professionalism	Section C Mandatory	
	- Worked, safely, individually, quietly, efficiently and/or with a high degree of professionalism	- Worked, safely, individually, quietly, efficiently and/or with considerable professionalism	- Somewhat worked, safely, individually, quietly, efficiently and/or with some professionalism	- Did not work, safely, individually, quietly, efficiently and/or with limited or no degree of professionalism	Section D Optional	
					Section E Mandatory	
					Section F Mandatory /Optional	

Category	5	3	1	0	Section	Score
Overall workmanship... Layout	- Structured and organized layout demonstrating forethought. - All sections and/or components placed to optimize the use of board space while maintaining professionalism.	- Acceptable structure and layout demonstrating some forethought. - Some sections and/or components placed to optimize the use of board space while maintaining professionalism.	- Layout is mostly unstructured and disorganized demonstrating little forethought. - Most sections and/or components placed do not optimize the use of board space and/or maintain professionalism.	- Totally unstructured and disorganized layout demonstrating no forethought. - No sections and/or components placed optimize the use of board space while maintaining professionalism.	All Sections	
Overall workmanship... Soldering	- No soldering or solder joint defects.	- Acceptable soldering and solder joints.	- 1-2 soldering or solder joint defects.	- More than 2 soldering or solder joint defects.	All Sections	
Overall workmanship... Placement	- All markings read in the same direction. - Components mounted flush to the board. - Components balanced between through holes.	- Markings are discernible. - Acceptable vertical mounting. - Acceptable horizontal mounting.	- 1-2 markings not discernible. - 1-2 tilted or poorly mounted components. - 1-2 poorly spaced or balanced components.	- Markings not discernible. - More than 2 tilted or poorly mounted components. - More than 2 poorly spaced or balanced components.	All Sections	
Overall workmanship... Leads	- Lead protrusion conforms to Class 3, 0.059”.	- Lead protrusion conforms to Class 2, 0.091”.	- 1-2 instances of excessive lead protrusion.	- More than 2 instances of excessive lead protrusion.	All Sections	
Overall workmanship... Stripping	- Insulation stripped ≤ 1 wire diameter.	- 1 incident of excessive wire insulation stripping.	- 2 incidents of excessive wire insulation stripping.	- More than 2 excessive wire insulation stripping.	All Sections	
Overall workmanship... Cleanliness and Damage	- All residues cleaned. - No damage to circuit, board or components.	- 1 incident of residue present. - 1 incident of observed damage.	- 2 incidents of residue present. - 2 incidents of observed damage.	- More than 2 incidents of residue present or not cleaned. - More than 2 incidents of observed damage.	All Sections	

Project Rubric – Section 2: Performance Characteristics60

Student Name: _____ Course/Section: _____ Date: _____

Category	<i>All must be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	Section	Score
Category	5	3	1	0		
Thinking, Communication, Knowledge and Understanding... Demonstrated in the calculations, simulations and data sheets.	- Demonstrates a thorough knowledge and understanding	- Demonstrates considerable knowledge and understanding	- Demonstrates some knowledge and understanding	- Demonstrates limited or no knowledge and understanding	Section A	
	- Uses conventions, vocabulary and terminology with a high degree of effectiveness	- Uses conventions, vocabulary and terminology with considerable effectiveness	- Uses conventions, vocabulary and terminology with some effectiveness	- Uses conventions, vocabulary and terminology with limited or no effectiveness	Section B	
	- Highly capable of applying scientific principles to solve technical problems effectively	- Considerably able to apply scientific principles to solve technical problems effectively	- Somewhat able to apply scientific principles to solve technical problems effectively	- Unable to apply scientific principles to solve technical problems effectively	Section C	
	- Highly capable of performing mathematics to solve problems effectively	- Considerably able to perform mathematics to solve problems effectively	- Somewhat able to perform mathematics to solve problems effectively	- Unable to perform mathematics to solve technical problems effectively	Section D	
					Section E	
					Section F	
Application... Demonstrated by taking and documenting measurements.	- Uses tools, equipment and materials with a high degree of professionalism and effectiveness	- Uses tools, equipment and materials with considerable professionalism and effectiveness	- Uses tools, equipment and materials with some professionalism and effectiveness	- Uses tools, equipment and materials with limited or no professionalism and effectiveness	Section A	
	- Able to follow both written and verbal instructions with a high degree of effectiveness and professionalism	- Able to follow both written and verbal instructions with considerable effectiveness or professionalism	- Able to follow both written and verbal instructions with some effectiveness or professionalism	- Able to follow both written and verbal instructions with limited or no effectiveness or professionalism	Section B	
	- Measures and documents appropriate data with a high degree of effectiveness and professionalism	- Measures and documents appropriate data with considerable effectiveness and professionalism	- Measures and documents appropriate data with some effectiveness and professionalism	- Measures and documents appropriate data with limited or no effectiveness and professionalism	Section C	
					Section D	
					Section E	
					Section F	

Project Rubric – Section 3: The Written Report60

Student Name: _____ Course/Section: _____ Date: _____

Category	<i>All must be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	<i>Any could be true to fall in this category and receive this score</i>	Score
Category	5	3	1	0	
Paper Formatting and Professional Appearance	- Strictly adhered to the “Overall Look and Feel”.	- Good adherence to the “Overall Look and Feel”.	- Poor adherence to the “Overall Look and Feel”.	- No adherence to the “Overall Look and Feel”.	
Category	3	2	1	0	
The Cover Page	- The title, the program code, the course code, the instructors name, your name, the creation date and the submission date are all present.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
Table of Contents	- All the sections of the report are listed sequentially and their pages are given.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
List of Figures and Tables	- All the figures and tables used in the report are listed sequentially and their pages are given.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
The Abstract	- Short, clear, complete and concise. - Key observations/ results and conclusions were listed and written in paragraph form.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
Introduction and Background	- The project, relevant background information, an indication of application or importance of the work is listed in paragraph form.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	

Objective	- The project objective is clearly detailed in a short paragraph.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
Timeline	- The timeline details the work, all major milestones, their dates and the duration of each entry from cradle to grave. - At least one timing image, figure or diagram is required for this section.	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
Category	6	4	2	0	
Equipment and BoM	- List of the material used in the experiment is provided - A BoM table detailing all the part numbers, descriptions, supplier names, quantities and pricing (in Canadian funds) is required. - Documented shipping costs and timing from each supplier	- One of the required data fields for this item is incorrect or not present.	- Two of the required data fields for this item is incorrect or not present.	- Three or more of the required data fields for this item is incorrect or not present.	
Category	8	4	2	0	
Procedure	- A detailed account of the procedure for this project completion is listed. - The procedural documentation is subcategorized for each major milestone and expressed systematically. - All the calculations performed for the design validation is included.	- One of the required data fields for this item is incomplete, incorrect or not present.	- Two of the required data fields for this item is incomplete, incorrect or not present.	- Three or more of the required data fields for this item is incomplete, incorrect or not present.	

	- At least one image, figure, simulation, sketch, schematic or wiring diagram is required.				
Observations / Analysis	- Presented all relevant experimental information both quantitative and qualitative forms. - Results may be expressed graphically in a graph or table. - Observations are also expressed textually with descriptive statements. - All calculations were performed in our course standard format.	- One of the required data fields for this item is incomplete, incorrect or not present.	- Two of the required data fields for this item is incomplete, incorrect or not present.	- Three or more of the required data fields for this item is incomplete, incorrect or not present.	
Category	6	4	2	0	
Conclusion and Recommendations (Lessons Learned)	- A well constructed, clearly and concisely stated conclusion supported with evidence from your observations and calculations. - The conclusion cited specific results, observations and calculations from the experiment are tied to the purpose. - Determined and summarized potential reasons for any discrepancies between measured results and the results predicted. - Potential sources of error and recommendations made with no use of catch-all phrases.	- One of the required data fields for this item is incomplete, incorrect or not present.	- Two of the required data fields for this item is incomplete, incorrect or not present.	- Three or more of the required data fields for this item is incomplete, incorrect or not present.	

Category	3	2	1	0	
Graphs, Tables, Figures and Images	- Graphs and tables are correct, complete, clear, logical and fully labeled. - Figure, table, image and graph captions and table titles accurate. - The data points should be drawn using the “curve of best fit”. - Graphs have titles, labels, coordinate labels, appropriate scale/axis units, etc. - Figures, tables and images are displayed in chronological order and labeled sequentially based on section.	- One of the required data fields for this item is incomplete, incorrect or not present.	- Two of the required data fields for this item is incomplete, incorrect or not present.	- Three or more of the required data fields for this item is incomplete, incorrect or not present.	
References	- All references used in your paper must be cited in MLS or APA format.	- One of the required data fields for this item is incomplete, incorrect or not present.	- Two of the required data fields for this item is incomplete, incorrect or not present.	- Three or more of the required data fields for this item is incomplete, incorrect or not present.	

I declare that I have performed all the calculations, prepared all the schematic diagrams, created all the simulations, took all the photos, performed all the measurements, recorded all the data, and prepared all aspects of this technical report alone and with no collaboration from anyone or anything (including the internet). If intellectual property from other authors was used (such as datasheet information and/or images, etc.), it is documented and referenced as such regardless if it was taken word for word or paraphrased.

I also declare that I have fully read and understood the project expectations and rubrics and have received clarification on anything I was unsure or unclear of. This includes, but is not limited to, what resources I may use to assist in any section of the project.

Printed Name

Signature

Date

Be sure to sign and date this document or no portion of your project will be graded.